14 An Introduction to Graphs: Finding Paths of Minimum Distance

This chapter covers

- Briefly introducing graphs from a theoretical point of view
- Strategies to implement graphs
- Formulating a problem: finding the best route for deliveries
- Search algorithms on graphs: BFS, DFS
- Using BFS to find the route that traverses the fewest blocks
- Minimum distance algorithms: Dijkstra – finding the shortest route
- Optimal search: A* algorithm (reading “A-star”) – finding the quickest route

We have already described the basics of trees in appendix C, and used several kinds of trees in the previous chapters: binary search trees, heaps, k-d trees etc...: you should be